Experiment 9

BI Security: Authentication, Authorization and Access Control using Microsoft Power BI

Aim: To understand and implement Business Intelligence security concepts including authentication, authorization, role-based access control, and data protection mechanisms using Microsoft Power BI.

OBJECTIVES:

1. To understand authentication methods in Business Intelligence systems.
2. To identify user roles and define access permissions.
3. To design role-based access control for organizational data.
4. To understand Row-Level Security (RLS) for restricting regional data access.
5. To identify security risks and recommend suitable security controls.

TOOLS / SOFTWARE REQUIRED:

1. **Operating System:** Windows 11.
2. **Software:** Microsoft Power BI Desktop.
3. **Dataset**: Global Superstore dataset in CSV or Excel format.
4. Power Query Editor
5. DAX (Data Analysis Expressions)
6. **System Requirements:** Minimum 4 GB RAM.

DATASET USED: Global Superstore Dataset.xlsx.

THEORY:

Business Intelligence (BI) security protects organizational data from unauthorized access and misuse. It ensures that users can access only the information permitted by their roles.

The major BI security concepts are:

1. **Authentication:** Verifies the identity of users through passwords, organizational accounts, and Multi-Factor Authentication (MFA).
2. **Authorization:** Determines the resources and operations a user is permitted to access.
3. **Role-Based Access Control (RBAC):** Assigns permissions according to organizational roles such as CEO, Sales Manager, Finance Manager, and Regional Manager.
4. **Row-Level Security (RLS):** Restricts users to viewing only the rows of data they are authorized to access.
5. **Data Protection:** Prevents unauthorized sharing, access, and leakage of sensitive business information.

PROCEDURE:

Part A: Security Role & Access Overview

1. Open the completed Global Superstore report in Microsoft Power BI Desktop.
2. Create a new report page by clicking the + (New Page) icon at the bottom of the Power BI window.
3. Rename the page to Security Role & Access Overview.
4. Create a new table using Home → Enter Data and name it `Role_Matrix`.
5. Add the following three columns:
    - Role
    - Information Requirements
    - Reports and Dashboards Access
6. Enter the following organizational role details:

Role	Information Requirements	Reports and Dashboards Access
CEO	Overall sales, profit, orders and business performance	All reports
Sales Manager	Daily sales, orders, products and customer performance	KPI Monitoring, Daily Sales, Monthly Executive
Finance Manager	Profit, profit margin and financial performance	KPI Monitoring, Weekly Profit, Monthly Executive
Regional Manager	Region-wise sales, profit and performance trends	KPI Monitoring, Monitoring Scenario, Daily Sales, Weekly Profit

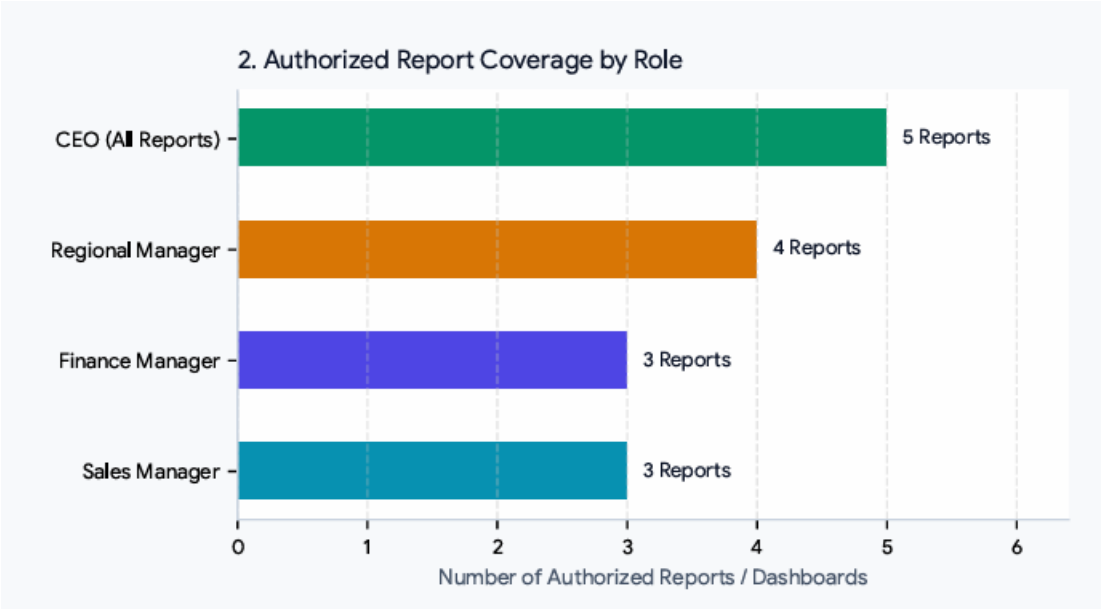
7. Click Load to add the table to the Power BI data model.
8. Select the Table visual from the Visualizations pane.
9. Drag the `Role`, `Information Requirements`, and `Reports and Dashboards Access` fields into the visual.
10. Resize the table and position it appropriately on the report canvas.
11. Open the Format visual pane, expand Title, enable it, and set the title to Organizational Role and Access Matrix.
12. Create another table using Home → Enter Data and name it `Access_Control`.
13. Add the following columns:
    - Role

- View
- Edit
- Share
- Admin

14. Enter the following permission details:

Role	View	Edit	Share	Admin
CEO	Yes	No	No	No
Sales Manager	Yes	No	No	No
Finance Manager	Yes	No	No	No
Regional Manager	Yes	No	No	No

15. Click Load to add the access control table.
16. Insert another Table visual into the report canvas.
17. Add the `Role`, `View`, `Edit`, `Share`, and `Admin` fields to the visual.
18. Set its title to Role-Based Access Control Matrix.
19. Insert a Text Box and add the following permission explanation:
    - CEO can view overall organizational reports.
    - Sales Manager can access sales-related reports.
    - Finance Manager can access financial reports.
    - Regional Manager can access regional performance reports.
    - BI Administrator retains editing, sharing, and administrative privileges.
20. Arrange both tables and the permission explanation neatly on the report page.
21. Apply suitable formatting, including readable fonts, table headers, background colors, and alignment.
22. Save the Power BI report.



Part B: Row-Level Security (RLS) Planning

1. Open the existing Power BI report and click the + (New Page) icon at the bottom of the window.
2. Rename the page to Row-Level Security (RLS) Planning.
3. Navigate to Home → Enter Data to create a new table.
4. Name the table `RLS_Design`.
5. Create the following four columns:
    - Role
    - Assigned Region
    - Data Access
    - RLS Filter Condition
6. Enter the following data:

Role	Assigned Region	Data Access	RLS Filter Condition
CEO	All Regions	All regional data	No regional restriction
Regional Manager – West	West	West region only	Region = West
Regional Manager – East	East	East region only	Region = East

Regional Manager – Central	Central	Central region only	Region = Central
Regional Manager – South	South	South region only	Region = South

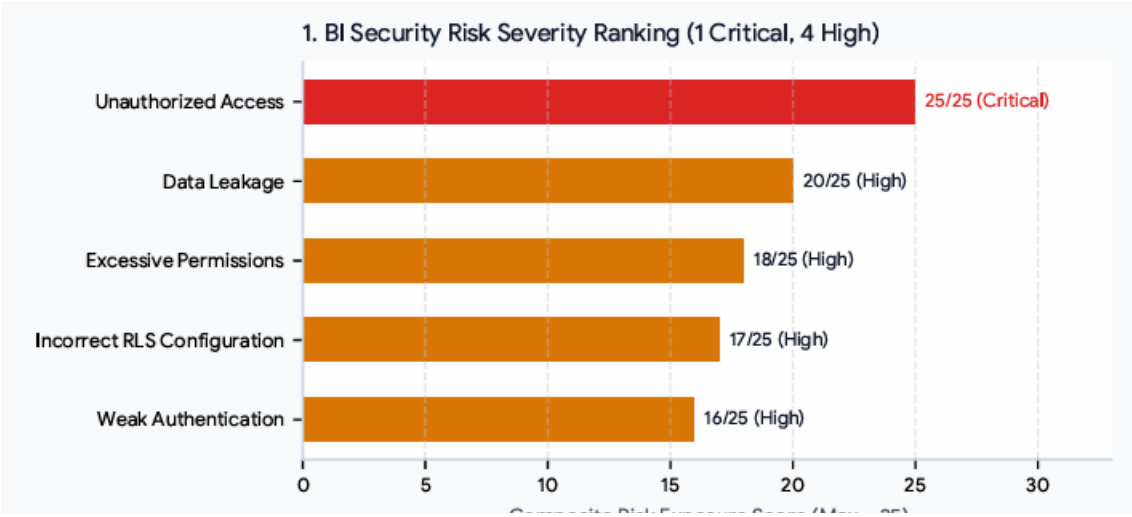
7. Click Load to add the table to the data model.
8. Select a Table visual from the Visualizations pane.
9. Add the `Role`, `Assigned Region`, `Data Access`, and `RLS Filter Condition` fields to the visual.
10. Set the visual title to Regional Row-Level Security Design Matrix.
11. Insert a Text Box and add the heading How Row-Level Security Improves Data Security.
12. Add the following points below the heading:
    - Restricts users to viewing only their assigned regional data.
    - Prevents unauthorized access to other regions.
    - Protects confidential sales and financial information.
    - Supports secure report sharing across departments.
    - Ensures users receive relevant business information
13. Insert three Text Boxes and format them as KPI-style cards.
14. Add the following information:

- 5 – Total RLS Roles
- 4 – Regional Restrictions
- All Regions – Executive Access

Create visualizations

15. Select a Donut Chart from the Visualizations pane.
16. Drag `RLS_Design[Assigned Region]` into Legend.
17. Drag `RLS_Design[Role]` into Values and change its aggregation to Count.
18. Set the chart title to Regional Access Distribution.
19. Insert a Clustered Bar Chart.
20. Drag `RLS_Design[Assigned Region]` into the Y-axis.
21. Drag `RLS_Design[Role]` into the X-axis and set its aggregation to Count.
22. Set the chart title to Number of Roles by Region.
23. Insert a Treemap visual.
24. Drag `RLS_Design[Data Access]` into Category.
25. Drag `RLS_Design[Role]` into Values and set its aggregation to Count.
26. Set the chart title to Regional Data Access Coverage.
27. Insert a Stacked Bar Chart.
28. Drag `RLS_Design[Assigned Region]` into the Y-axis.
29. Drag `RLS_Design[Role]` into the X-axis and set its aggregation to Count.
30. Drag `RLS_Design[Data Access]` into Legend.
31. Set the chart title to Regional Access Classification.
32. Insert another Table visual and add `Role`, `Assigned Region`, and `Data Access`.
33. Set its title to Regional Security Role Summary.

34. Arrange all tables, cards, charts, and text boxes neatly on the page.
35. Apply suitable colors, titles, labels, and formatting to make the report easy to understand.
36. Save the Power BI report.



Part C: Security Risk Assessment Dashboard

1. Open the existing Power BI report and click the + (New Page) icon at the bottom of the window.
2. Rename the page to Security Risk Assessment Dashboard.
3. Navigate to Home → Enter Data to create a new table.
4. Name the table `Security_Risks`.
5. Create the following five columns:
   - Security Risk
   - Likelihood
   - Impact
   - Risk Level
   - Mitigation
6. Enter the following data:

Security Risk	Likelihood	Impact	Risk Level	Mitigation
Unauthorized Access	High	High	Critical	Implement authentication and MFA
Excessive Permissions	Medium	High	High	Apply least privilege access

Data Leakage	Medium	High	High	Restrict sharing and export permissions
Weak Authentication	Medium	High	High	Enforce strong passwords and MFA
Incorrect RLS Configuration	Medium	High	High	Test and validate RLS roles

7. Click Load to add the table to the data model.
8. Select a Table visual from the Visualizations pane.
9. Add all five columns to the visual.
10. Set the visual title to Organizational BI Security Risk Assessment.

Apply conditional formatting

11. Select the table visual.
12. Open the Format visual pane.
13. Locate the `Risk Level` column under the cell elements or conditional formatting options.
14. Enable background color formatting and configure the following colors:
    - Critical – Red
    - High – Orange
    - Medium – Yellow
    - Low – Green
15. Apply the formatting to make the different risk levels easy to identify.

Create summary cards

16. Insert three Text Boxes and format them as KPI-style cards.
17. Add the following information:
    - 5 – Total Security Risks
    - 1 – Critical Risks
    - 4 – High Risks

Create visualizations

18. Insert a Donut Chart.
19. Drag `Security_Risks[Risk Level]` into Legend.
20. Drag `Security_Risks[Security Risk]` into Values and set the aggregation to Count.
21. Set the chart title to Security Risk Distribution.
22. Insert a Clustered Bar Chart.
23. Drag `Security_Risks[Security Risk]` into the Y-axis.
24. Drag `Security_Risks[Security Risk]` into the X-axis and set the aggregation to Count.
25. Drag `Security_Risks[Impact]` into Legend.

26. Set the chart title to Security Risks by Impact.
27. Insert a Treemap visual.
28. Drag `Security_Risks[Mitigation]` into Category.
29. Drag `Security_Risks[Security Risk]` into Values and set the aggregation to Count.
30. Set the chart title to Risk Mitigation Overview.

Add security recommendations

31. Insert a Text Box with the heading Recommended Security Controls.
32. Add the following recommendations:
    - Unauthorized Access: Implement Multi-Factor Authentication (MFA).
    - Excessive Permissions: Apply the principle of least privilege.
    - Data Leakage: Restrict sharing and data exports.
    - Weak Authentication: Enforce strong authentication policies.
    - Incorrect RLS Configuration: Regularly test and validate security roles.
33. Arrange the table, summary cards, charts, and recommendations neatly on the report canvas.
34. Apply suitable colors, labels, titles, and formatting.
35. Save the Power BI report.

1. Regional Row-Level Security (RLS) DAX Filter Matrix

Role	Assigned Region	Data Access	RLS DAX Filter
CEO	All Regions	All regional data	No regional restriction
Regional Manager - West	West	West region only	[Region] = "Western US"
Regional Manager - East	East	East region only	[Region] = "Eastern US"
Regional Manager - Central	Central	Central region only	[Region] = "Central US"
Regional Manager - South	South	South region only	[Region] = "Southern US"

Part D: Authentication & Security Recommendations

1. Open the existing Power BI report and click the + (New Page) icon at the bottom of the window.
2. Rename the page to Authentication & Security Recommendations.
3. Navigate to the existing `Authentication_Methods` table. If it has not already been created, use Home → Enter Data to create it.
4. Create the following three columns:
    - Authentication Method
    - Description
    - Security Advantages
5. Enter the following data:

Authentication Method	Description	Security Advantages
Username-Password	Users authenticate using a username and password.	Provides basic account protection
Organizational Account	Users sign in using their organization's Microsoft account.	Supports centralized identity and access management
Multi-Factor Authentication (MFA)	Users verify their identity using an additional authentication factor.	Adds an extra layer of account security.

6. Click Load to add the table if it is newly created.
7. Insert a Table visual and add all three fields.
8. Set its title to Authentication Methods Comparison.

Create summary cards

9. Insert three Text Boxes and format them as KPI-style cards.
10. Add the following information:
    - 3 – Authentication Methods
    - Organization – Centralized Identity
    - MFA – Additional Security Layer

Create the security recommendations table

11. Navigate to Home → Enter Data.
12. Create a table named `Security_Recommendations`.
13. Add the following four columns:
    - Security Issue
    - Recommended Control
    - Priority
    - Expected Outcome
14. Enter the following data:

Security Issue	Recommended Control	Priority	Expected Outcome
Unauthorized Access	Implement MFA	Critical	Stronger user verification

Excessive Permissions	Apply Least Privilege Access	High	Restricted unauthorized actions
Data Leakage	Restrict Sharing and Exporting	High	Improved data protection
Weak Authentication	Enforce Strong Authentication	High	Reduced account security risks
Incorrect RLS Configuration	Regularly Test RLS Roles	High	Accurate regional data restrictions

15. Click Load to add the table.
16. Insert a Table visual and add all four fields.
17. Set the visual title to Security Recommendations Matrix.

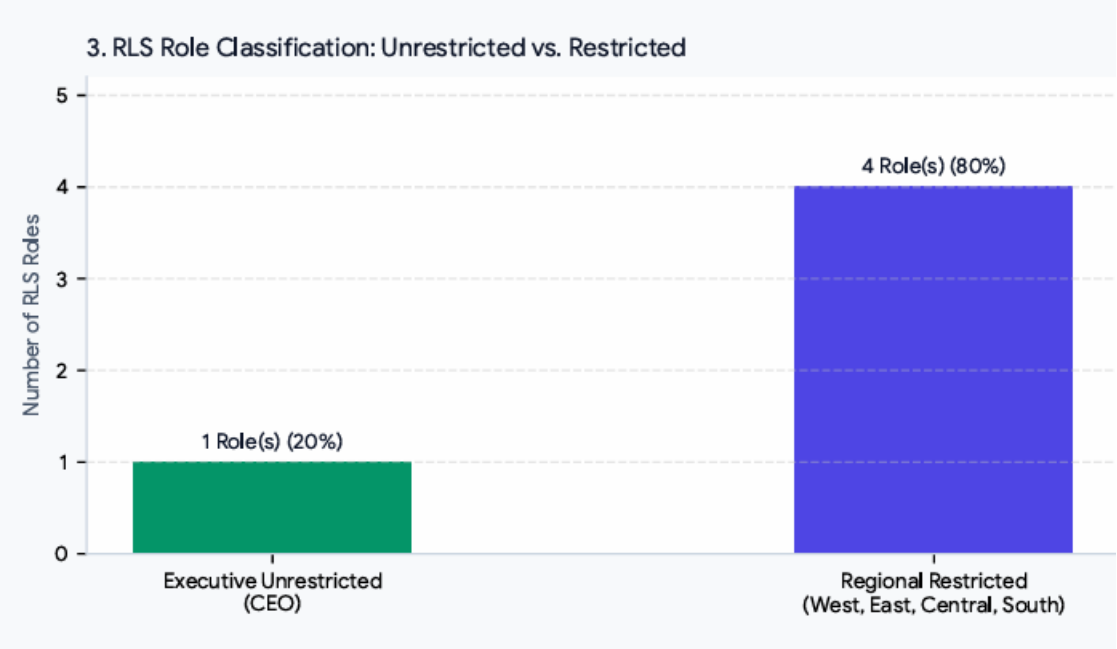
Apply conditional formatting

18. Select the recommendations table.
19. Open the Format visual pane.
20. Locate the `Priority` column's conditional formatting settings.
21. Apply the following background colors:
    - Critical – Red
    - High – Orange
    - Medium – Yellow
    - Low – Green

Create visualizations

22. Insert a Donut Chart.
23. Drag `Security_Recommendations[Priority]` into Legend.
24. Drag `Security_Recommendations[Security Issue]` into Values and select Count.
25. Set the title to Security Recommendations by Priority.
26. Insert a Clustered Bar Chart.
27. Drag `Security_Recommendations[Security Issue]` into the Y-axis.
28. Drag `Security_Recommendations[Security Issue]` into the X-axis and select Count.
29. Drag `Security_Recommendations[Priority]` into Legend.
30. Set the title to Security Issues by Priority.
31. Insert a Treemap visual.
32. Drag `Security_Recommendations[Recommended Control]` into Category.
33. Drag `Security_Recommendations[Security Issue]` into Values and select Count.
34. Set the title to Security Controls Overview.
35. Insert a Clustered Column Chart.

36. Drag `Authentication_Methods[Authentication Method]` into the X-axis.
37. Drag `Authentication_Methods[Authentication Method]` into the Y-axis and select Count.
38. Set the title to Authentication Methods Comparison.



Result

The BI Security report was successfully prepared using Microsoft Power BI. It includes organizational role and access matrices, Row-Level Security (RLS) planning, security risk assessment, authentication method comparison, and security recommendations. Various tables, cards, donut charts, bar charts, and treemaps were used to present the security analysis.

LEARNING OUTCOMES:

- Understood the concepts of authentication and authorization in Business Intelligence.
- Learned to design role-based access control and permission matrices.
- Understood the importance of Row-Level Security (RLS) in protecting regional data.
- Identified potential BI security risks and suitable mitigation techniques.
- Gained practical experience in creating security analysis dashboards using Microsoft Power BI.